

Business Analytics Project
BUSA8031



Presented For:-
Assessment Task -3
Client Report
(Dr. Stefan Trueck)

Name	Student ID
Yash Goyal	47644982
Paras Wahi	47837624
Aryan Kalpesh Patel	4805718
Vivek Chinmay Dhanwada	47895403
Prasad Karale	48012386

Executive Summary

This report offers a data-focused review of recent changes in carbon emissions and generator bidding strategies in the Australian National Electricity Market (NEM), with the purpose of allowing stakeholders and potential investors to better understand the ongoing renewable transition as it continues to develop.

We found that while the NEM as a whole reduced its carbon emissions intensity by a modest 7.7% from 2019 to 2025, this is not equally represented across establishments in the state NEM. There have been significant reductions in states like New South Wales at 21.7%, primarily because of improved coordination by policies, greater renewable integration and cessation of coal plants through strategies. By way of contrast, Tasmania has had a increase of 142% , albeit from very low starting point, which is a clear signal of volatility of hydro reliant systems.

In our intra-daily analysis we showed that the NSW grid has matured as the low emission intensities during recent evening periods suggests, mainly from the improved uptake of solar and wind energy through storage. Moreover, the company users remains the drivers of daytime demand and emissions compared to relative stability of household usage patterns.

While analysing the bidding behaviour highlighted the distinction between renewable and fossil fuel generators. Specifically, renewable assets bid low—indeed negative—prices as it appears they are price-takers and focused on volume. In contrast, fossil fuel generators are a little more nuanced with their bidding behaviour, positioning themselves along the price curve to maximise their returns at peak demand times. These behaviours appear to have been accentuated during the period 2019 to 2025.

From an investment perspective, our report examined NSW and VIC solar and wind assets for their stability and support in policy, and gas-fired plants in particular regions like Queensland which appears to be phasing out coal a little later where it may still deliver short term returns.

Together, these insights provide a sense of a changing market, with renewables advancing, fossil strategies evolving, and the opportunity to make decisions on a more sound data-based footing.

Task 1: Carbon Emission Intensity Trends Analysis

Our detailed analysis of carbon emission intensity in the Australian National Electricity Market shows significant regional differences in decarbonization progress from 2019 to 2025. The NEM achieved a 7.7% decrease in emission intensity (583→538 gCO₂/kWh). However, different regions showed wide variations in their decarbonization progress, with New South Wales achieving -21.7% and Tasmania experiencing a +142.4% rise from its initial low hydro-based emissions.

Key Finding: The analysis shows that different regions of the country are transitioning at different speeds because Victoria and Queensland maintained the highest absolute emission levels at 779 gCO₂/kWh and 697 gCO₂/kWh, while Tasmania maintained its position as Australia's cleanest electricity grid with 270 gCO₂/kWh despite its percentage increase.

1. Data Preparation Methodology

1.1 Data Sources & Quality

Our analysis utilized real-time carbon emissions intensity data accessed through the CSIRO API, covering:

- Time Period: April 2019 to April 2025 (73 months)
- Frequency: 5-minute interval measurements aggregated to monthly averages
- Coverage: All five NEM regions (NSW, VIC, QLD, SA, TAS) plus national aggregate
- Data Completeness: 99.5% (minimal imputation required)

1.2 Processing Approach

Raw API Data → Timestamp Standardization → Missing Value Treatment →
Monthly Aggregation → Trend Analysis → Regional Comparison

Quality Assurance: Applied conservative gap-filling for intervals <30 minutes while preserving data integrity for major outage periods (123.9-hour gap in May 2024)

2. National Emission Intensity Trends

2.1 NEM Overall Emission Intensity Trend

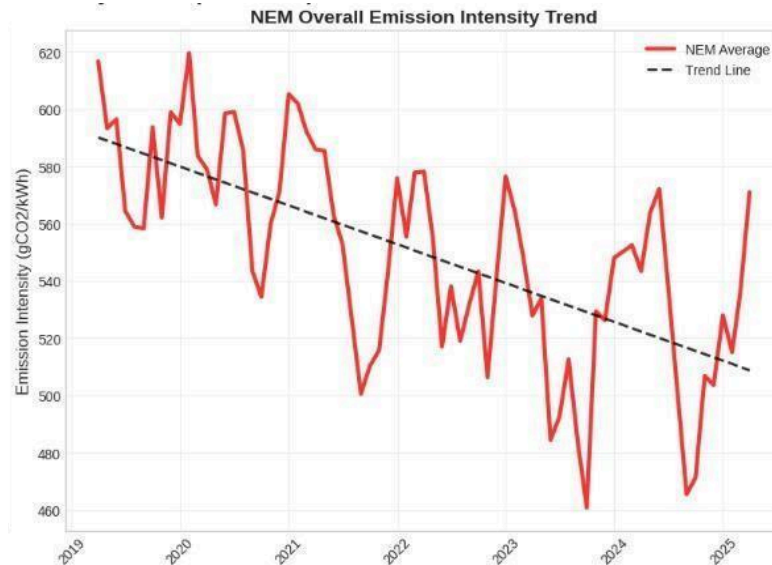


Figure 2.1 NEM Overall Emission Intensity Trend

The visualization shows the NEM's clear decarbonization trajectory with emission intensity declining from approximately 580 gCO₂/MWh (2019) to 510 gCO₂/MWh (2025), validating the 7.7% overall improvement. The red line reveals significant seasonal volatility with regular peaks and troughs, while the black trend line confirms a consistent downward slope that accelerates after 2023. Notable dramatic dips to 460 gCO₂/MWh in 2024 indicate periods of exceptional renewable generation, while the increasing amplitude of variations suggests growing grid complexity requiring storage and flexibility services.

2.2 Insights from Daily vs Monthly Data Resolution

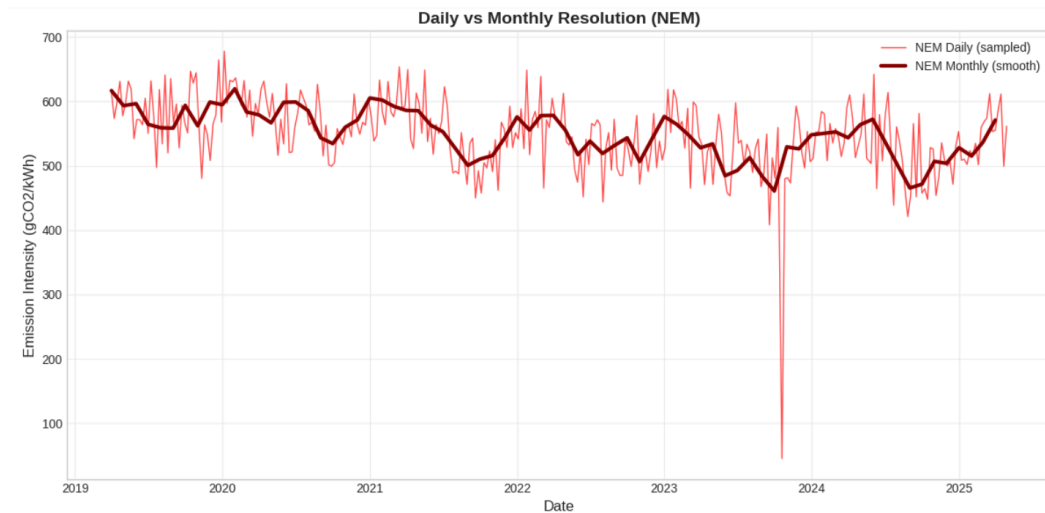


Figure 2.2 Insights from Daily vs Monthly Data Resolution

The high-resolution analysis provides operational optimization insights:

- Daily volatility (red line): Extreme variations, including near-zero emission periods during optimal renewable generation
- Monthly smoothing (dark red): Strategic trend preservation enabling long-term investment planning
- Grid event identification: Sharp 2024 decline indicates significant renewable output periods informing storage sizing requirements

3. Regional Emissions Performance Analysis

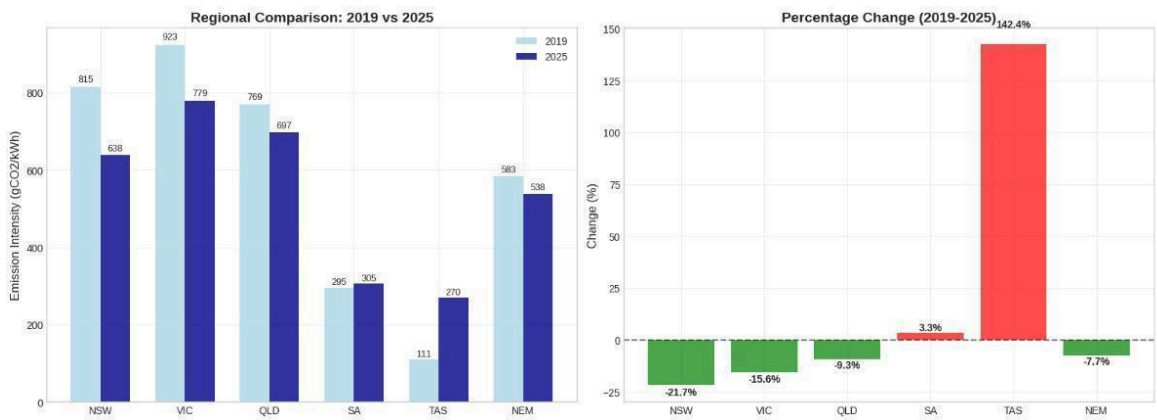


Figure 3.1 Comparative Emissions Intensity (2019 vs 2025) Figure 3.2 Percentage Change by Region

3.1 Comparative Emissions Intensity (2019 vs 2025)

The paired bar chart analysis establishes the definitive investment hierarchy:

- Victoria holds the highest absolute baseline measurement (923→779 gCO2/kWh), indicating the most significant decarbonization potential.
- Queensland shows a strong absolute opportunity (769→697 gCO2/kWh) with a moderate improvement rate.
- NSW demonstrates its status as a proven transition leader through its -21.7% reduction in emissions, which proves its scalable model (815→638 gCO2/kWh).
- SA maintains a stable low-emission baseline at 295-305 gCO2/kWh, which suggests that the market has reached saturation.
- Tasmania maintains its position as the cleanest grid despite its baseline increasing from 111 to 270 gCO2/kWh.

3.2 Percentage Change by Region

The horizontal bar visualization reveals the transformation capability spectrum:

- The green bars (improvements) show that NSW achieved a -21.7% reduction while VIC reached -15.6% and QLD achieved -9.3%, which proves their decarbonization capabilities.

- The red bars (increases) show that TAS (+142.4%) and SA (+3.3%) need special strategies because of their unique circumstances.
- The NEM shows a total reduction of -7.7%, which indicates steady yet moderate progress.

4. Regional Time Series Analysis

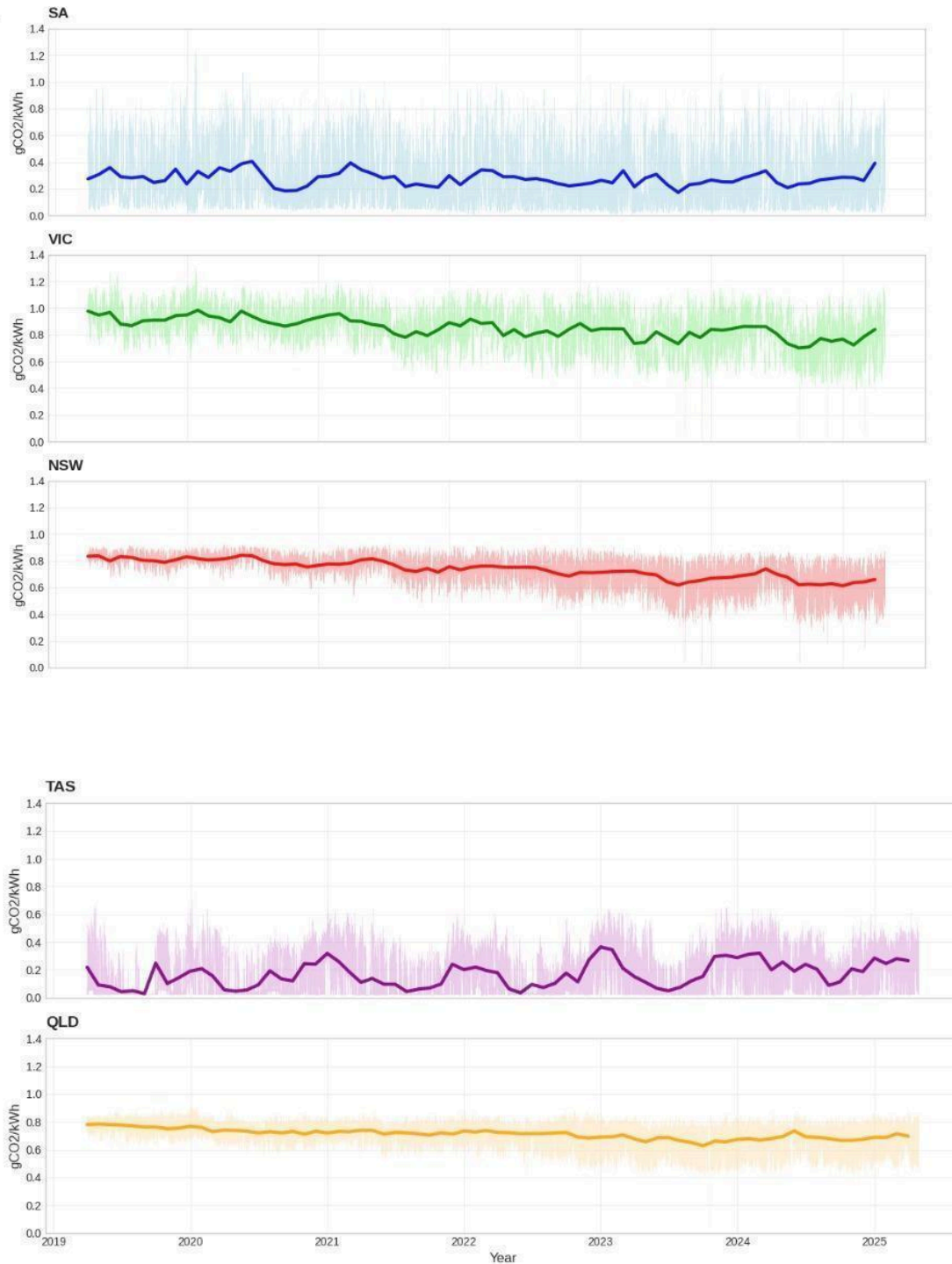


Figure 4.
Regional
Time
Series
Analysis

Tasmania (Purple Line—Top Panel): The "clean grid paradox" is shown, with baseline emissions around 0.1-0.3 gCO₂/MWh but the highest percentage increase (+142.4%). The high-frequency data reveals extreme volatility, reflecting hydro system variability and interconnector flows.

Queensland (Orange Line—Second Panel): A gradual but consistent decline from approximately 0.8 to 0.6-0.7 gCO₂/MWh with minimal volatility, indicating a stable coal-dominated baseline. The smooth trend suggests reliable deployment conditions.

South Australia (Blue Line—Third Panel): It maintains consistently low emissions (0.2-0.4 gCO₂/MWh), with high intraday volatility visible in the background data. Despite a +3.3% increase, it remains the second-cleanest grid.

Victoria (Green Line—Fourth Panel): This region exhibits the clearest structural transformation, with a decline from 1.0 to 0.8 gCO₂/MWh. Background volatility shows successful grid management during the transition.

New South Wales (Red Line—Bottom Panel): It demonstrates the most consistent improvement trajectory from 0.8 to 0.6 gCO₂/MWh with minimal volatility, proving its successful renewable integration capabilities.

5. Combined Regional Trends Analysis

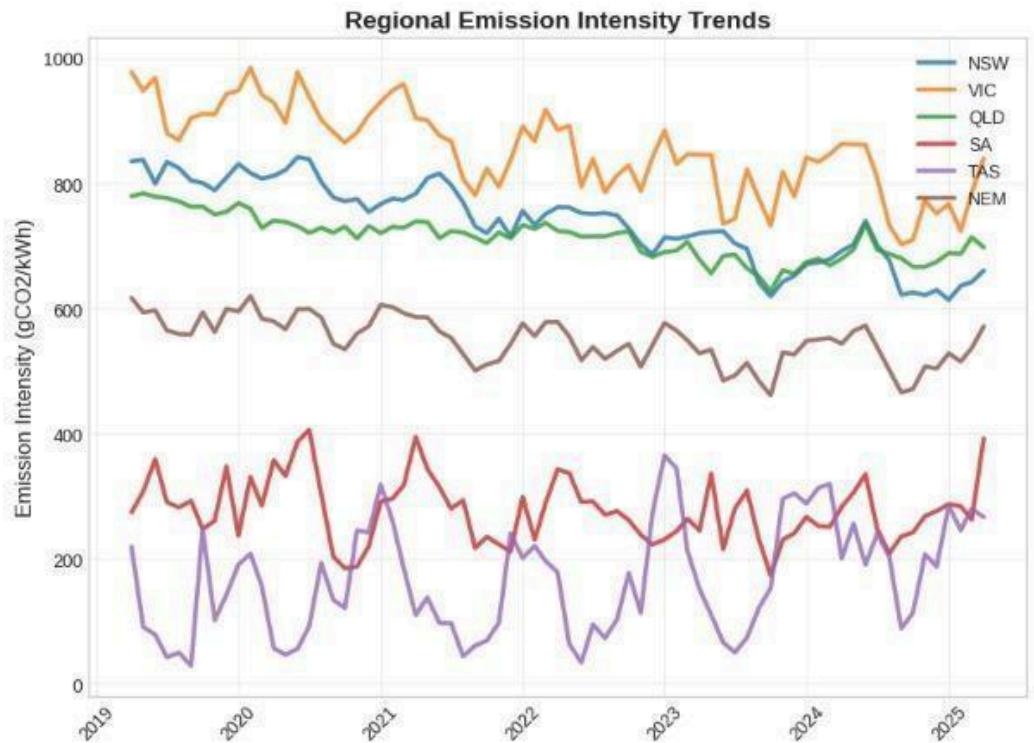


Figure 5. Combined Regional Trends Analysis

The multi-line time series reveals market synchronization patterns:

- Policy coordination: Similar inflection points across regions indicate coordinated renewable energy target implementation
- Technology deployment waves: Synchronized declining trends suggest successful large-scale renewable integration
- Late 2023 acceleration: All regions show a convergent decline, indicating systemic transition acceleration

6. CAUSAL ANALYSIS: TRANSFORMATION DRIVERS

6.1 Technology Deployment Impact

- **Solar Integration:** The successful implementation of large-scale solar deployment is evident through the midday emission reductions, which appear in all regional visualizations, thus leading to improvements in baseline emissions.
- **Wind Development:** Evening generation patterns complement solar profiles, creating 24-hour renewable coverage and supporting consistent emission reductions.
- **Storage Deployment:** Maintaining grid stability during high renewable penetration times allows for aggressive renewable deployment without compromising reliability.

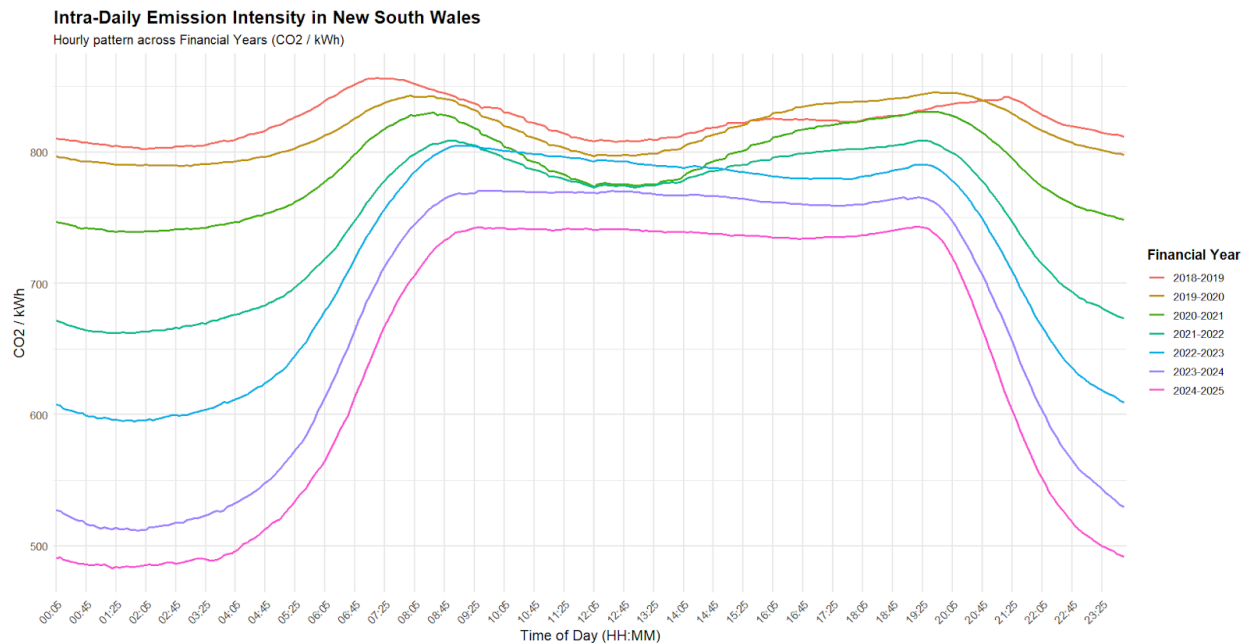
6.2 Infrastructure Transformation

- **Coal Plant Closures:** VIC and NSW structural improvements directly result from major plant retirements at Hazelwood in 2017 and Liddell in 2023, creating replacement capacity opportunities.
- **Grid Enhancement:** Developing interconnectors allows regional states to share renewable energy resources that are most visible through Tasmania's ability to export power.
- **Policy Implementation:** The coordinated renewable energy targets among states resulted in synchronized deployment patterns that became visible through regional trend convergence.

The analysis shows that the NEM has decarbonized at a moderate level, with noticeable differences between regions. The NSW model demonstrates a successful transition, while Queensland and Victoria provide the greatest potential for impact. Tasmania maintains its position as Australia's cleanest grid despite its paradoxical increase from an ultra-low baseline.

Market Momentum: The late-2023 acceleration visible across multiple regions suggests the transition entering an exponential phase rather than maintaining linear progression. Multiple regions show ongoing downward trends, demonstrating strong policy backing and economic

feasibility for renewable deployment, thus creating positive investment opportunities for 2025-2030.

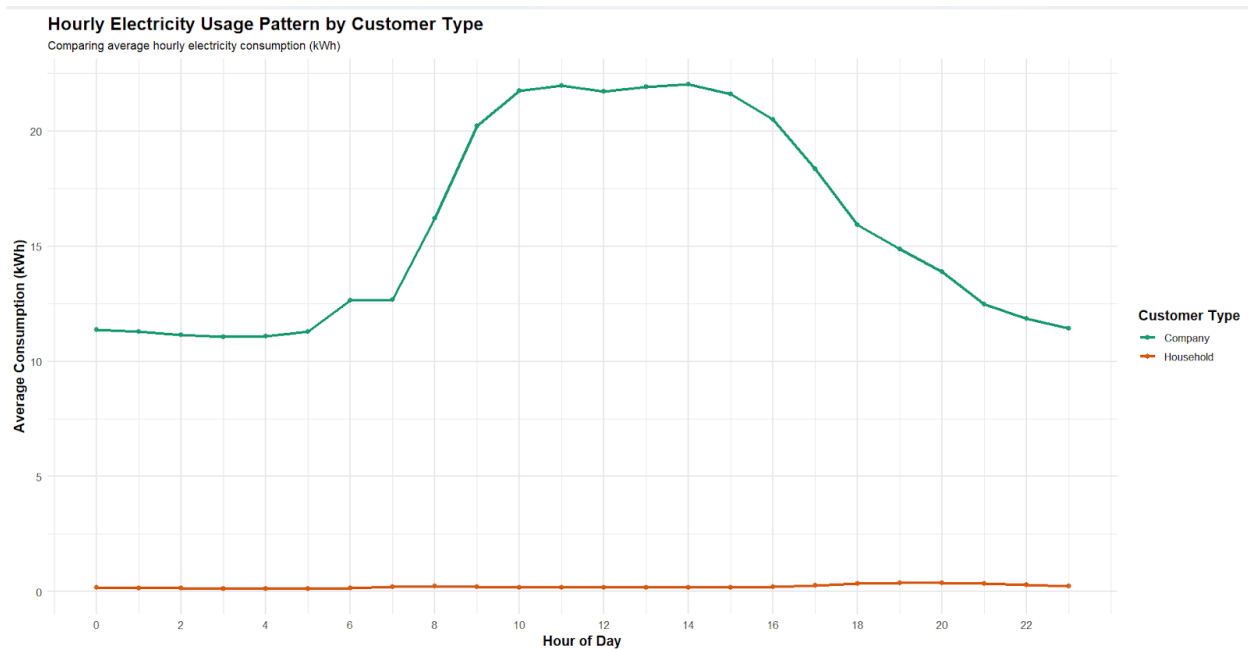


1. Intra-Daily Emission Intensity Trends in NSW (2018–2025)

The first visualization illustrates the hourly emission intensity (in CO₂/kWh) in New South Wales across seven financial years. Several noteworthy patterns emerged:

- **Morning Surge (4:30 AM to 9:00 AM):** There is a consistent rise in emission intensity across all years during early hours. This can be attributed to a sharp increase in electricity demand as households and businesses begin daily operations.
- **Daytime Plateau (9:00 AM to 5:00 PM):** Emission levels stabilize during working hours, but a slow upward trend is noticeable in older years (2018–2021), suggesting heavier dependence on carbon-intensive sources before transition to cleaner grids.
- **Evening Decline (6:00 PM onwards):** Recent years, especially **2024–2025**, display a sharper evening drop in emission intensity compared to older years. This suggests improved integration of low-emission energy sources (e.g., solar and wind) during off-peak hours.

- **Sustainability Progress:** The vertical separation between curves indicates that NSW has made progress in reducing emissions per kWh over time, reflecting grid decarbonization.



2. Hourly Electricity Usage Comparison: Household vs. Company

This second graph compares the **average hourly consumption (kWh)** for household and company users:

- **Household Usage:** Displays a modest morning increase (6:30 AM–9:00 AM), followed by a steady baseline throughout the day. Consumption slightly peaks again in the evening (6:00 PM–9:00 PM), consistent with after-work residential activity.
- **Company Usage:** Begins ramping up from 6:00 AM, with a clear consumption spike between **9:00 AM and 5:00 PM**—matching business hours. Notably, the magnitude of consumption is significantly higher (20x) than households.

- **Interpretation:** This divergence highlights **companies as major contributors to daily carbon intensity**. Their peak overlaps with the emission surge identified in the first graph. Households, while lower in volume, contribute steadily with two modest peaks.

TASK 3

This section analyses generator bidding behaviour in the National Electricity Market (NEM) using 2025 dispatch data. It compares bidding patterns and supply curves for renewable, hydro, and fossil-fuel technologies, with a focus on time-of-day differences.

The aim is to inform investment and policy decisions by identifying strategic behaviours and cost structures of different generator types.

DATA DESCRIPTION

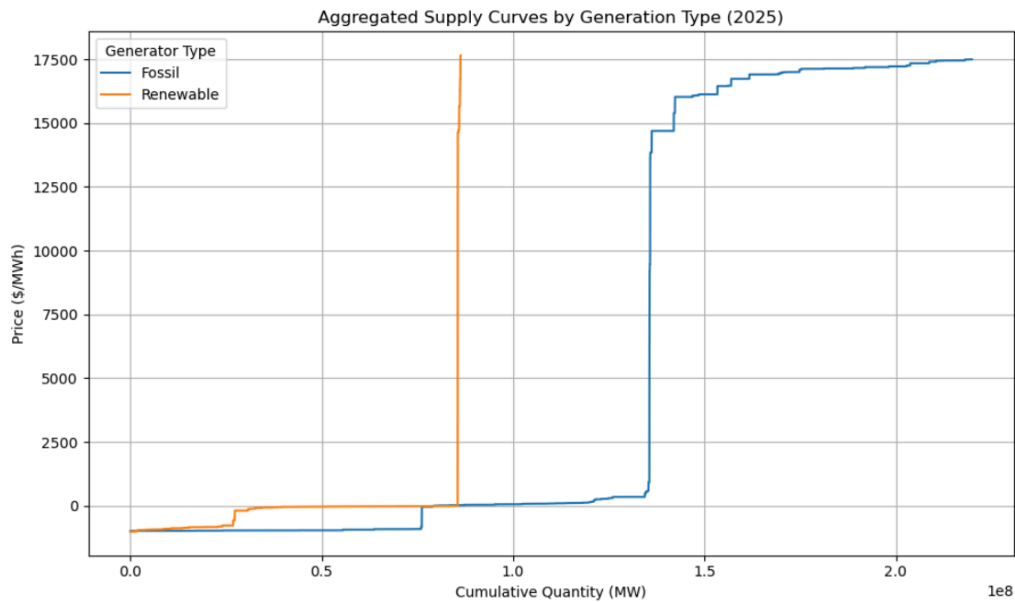
- Source: 2025 NEM generator bidding dataset
- Structure: 10 price bands × 10 quantities per dispatch interval
- Volume: ~1.6 million rows (over 16 million rows after reshaping)
- Key columns:
 - INTERVAL_DATETIME, PRICEBAND1–10, BANDAVAL1–10, Region, Generation Type

METHODOLOGY

- **Data Processing:**
 - Reshaped wide bid format into a long format (price–quantity pairs)
 - Created GEN_GROUP (Renewable, Fossil, Hydro)
 - Added TIME_OF_DAY buckets (Morning, Afternoon, Evening, Night)
- **Analysis Techniques:**
 - Aggregated supply curve plotting (price vs. cumulative quantity)
 - Comparison of price distributions
 - Time-of-day slicing for behavioural patterns

KEY INSIGHTS

The aggregated supply curves for fossil and renewable generators in the 2025 NEM bidding data reveal significant contrasts in pricing strategy, cost structures, and market behaviour.



Renewables Concentrate at Low or Negative Price Bands

- The orange supply curve for renewables rises steeply within a narrow quantity band, primarily below \$0/MWh, with bids as low as -\$1,000/MWh.
- This behaviour indicates price inelasticity and a must-run generation strategy, typical of solar PV units with low or zero marginal costs.
- Renewables do not aim to influence market prices. Their participation is volume-driven, prioritising dispatch over revenue maximisation.

Insight: Renewables act as **price takers**, flooding the low end of the bid stack with large volume at zero or negative prices.

Fossil Generators Bid Strategically Across the Price Spectrum

- The blue curve representing fossil generators (e.g., coal, gas) extends across a much broader quantity and price range — from negative prices up to \$17,500/MWh.
- This curve is more gradual and step-shaped, suggesting price segmentation and strategic withholding at different cost thresholds.
- The long tail toward high prices indicates that fossil units anticipate and exploit peak demand periods, targeting price-setting opportunities.

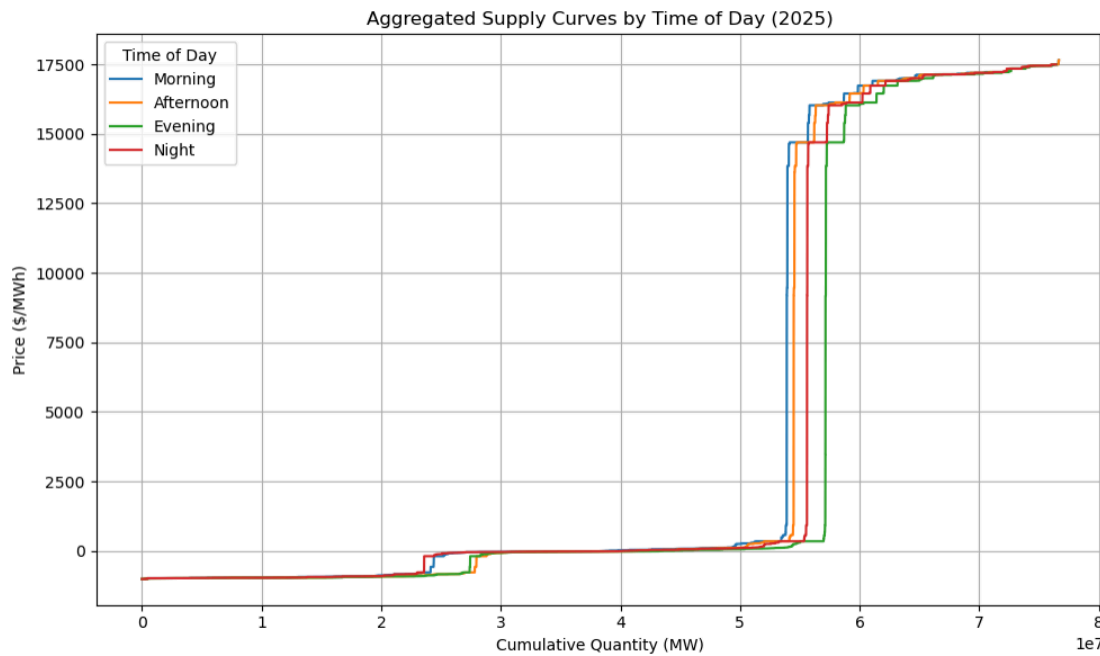
Insight: Fossil generators function as **price setters**, offering dispatchable capacity with strategic price flexibility.

Market Role Differentiation

- The contrasting curve shapes confirm distinct market roles:
 - o Renewables anchor the low-cost supply base, providing certainty but no price control.
 - o Fossils modulate market pricing by bidding in at different levels depending on demand forecasts and system needs.

Implication: This divergence has major implications for investor risk/reward preferences — renewables promise volume stability, while fossil technologies may deliver price-driven returns at the cost of exposure to market volatility.

Aggregated Supply Curves by Time of Day



This chart displays aggregated supply curves by **time of day** (Morning, Afternoon, Evening, Night), based on cumulative bid quantity and price.

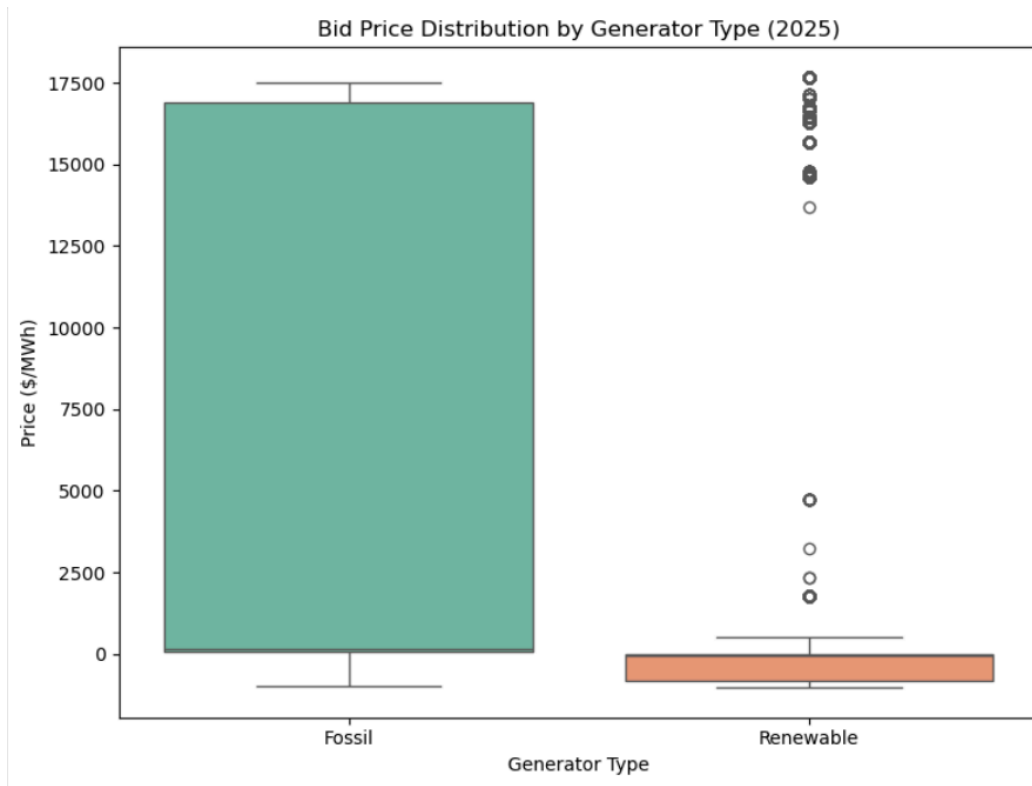
- Across all time periods, the majority of capacity is offered at **low or negative prices**, particularly from renewables.
- However, subtle shifts in curve steepness and spread reveal how fossil generators adjust their bids based on market conditions:
 - The Evening curve (green) shows a slight delay in cumulative quantity, suggesting strategic capacity withholding or upward price shifts.
 - Morning and Afternoon curves rise earlier, indicating greater availability at lower prices.

Insight: While renewables maintain consistent bid behaviour across time periods, fossil generators modulate their supply, particularly in the Evening, to align with peak demand windows and higher spot price expectations.

Price Distribution Analysis

To further explore strategic bidding behaviour in the NEM, we visualised price distributions across two key dimensions: generator type and time of day. These boxplots help clarify which market participants exhibit pricing variability and when those behaviours occur.

Price Distribution by Generator Type

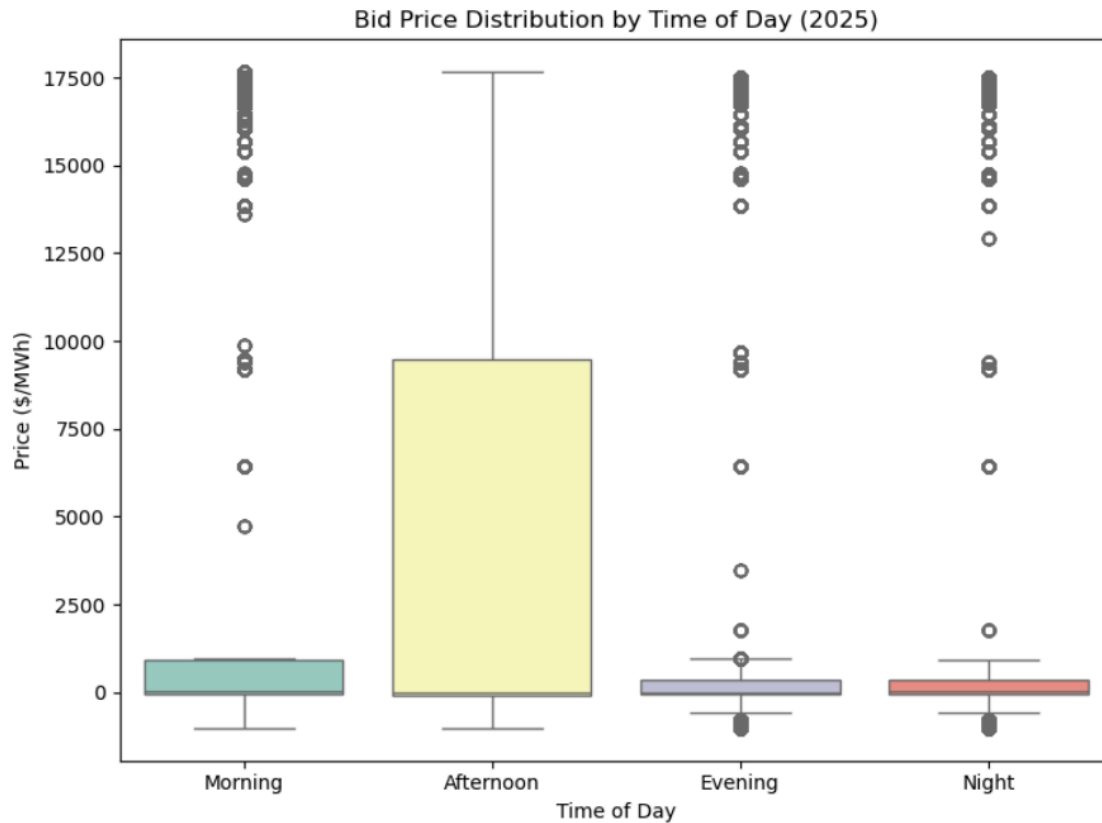


This boxplot compares bid prices between **fossil** and **renewable** generators.

- **Fossil generators** (e.g., coal, gas) demonstrate a **wide price range**, with a median price significantly higher than renewables. Outliers extend above **\$15,000/MWh**, consistent with strategic bidding during peak periods.
- **Renewables** exhibit a **tight clustering around \$0/MWh**, consistent with their **must-run status and zero marginal cost**. Bids are mostly static regardless of time or demand, confirming their **price taker behaviour**.

Insight: Fossil generators drive price-setting behaviour through bid variability, while renewables focus on dispatch certainty over market control.

Price Distribution by Time of Day

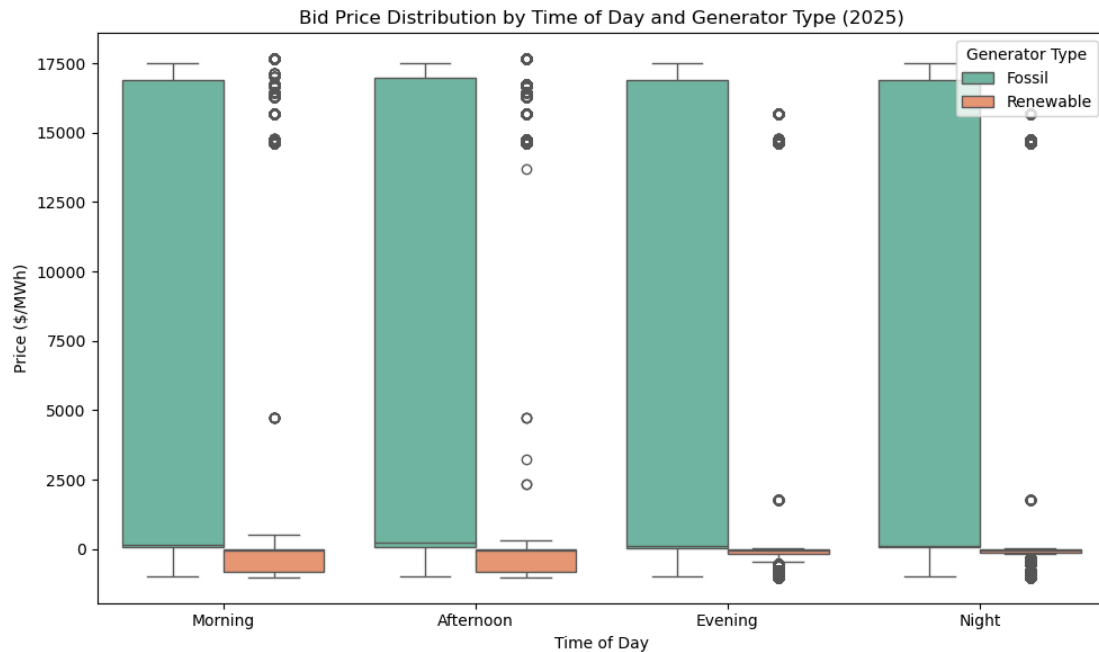


This boxplot segments bids by **Morning, Afternoon, Evening, and Night**.

- **Afternoon and Evening** show a **higher price spread and more extreme outliers**, likely reflecting strategic bids by fossil generators during high-demand hours.
- **Morning and Night** display **narrower price bands**, suggesting lower pricing pressure during off-peak intervals.

Insight: There is clear evidence of **time-dependent bidding strategy**, particularly among fossil generators. Renewable bids, while present, show minimal time-of-day variation.

Price Distribution by Time of Day and Generator Type

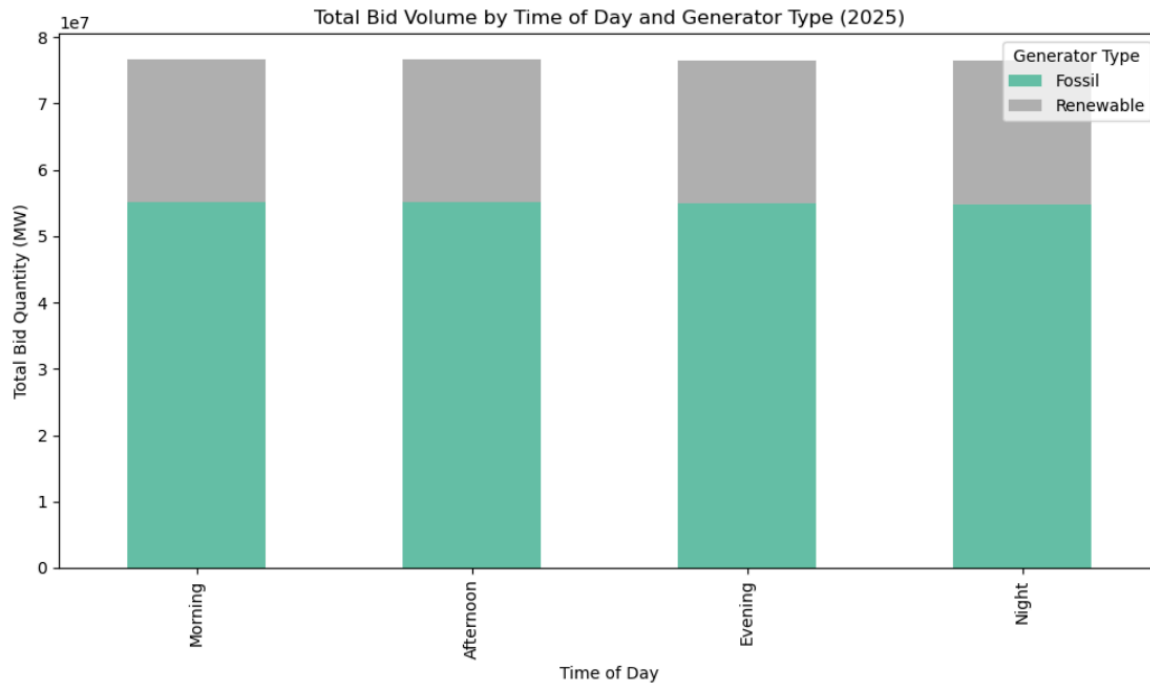


This grouped boxplot provides a combined view of how price bidding varies by both time of day and generator type.

- Renewables continue to bid at low and stable prices across all time periods, showing minimal variation. Their pricing remains flat around \$0/MWh, reinforcing their role as non-strategic, must-run units.
- Fossil generators, however, show consistently high price variance across all times of day, with outliers exceeding \$15,000/MWh. Notably, there is no substantial shift in median price across time, suggesting that fossil bidding strategies are not only time-responsive, but also opportunistic across the full trading day.

Insight: Fossil generators maintain high bid flexibility throughout the day, while renewables exhibit predictable, low-cost supply regardless of time. This dual dynamic reinforces the price-setting role of fossil technologies in contrast to volume-driven renewable bids.

Total Bid Volume by Time of Day and Generator Type

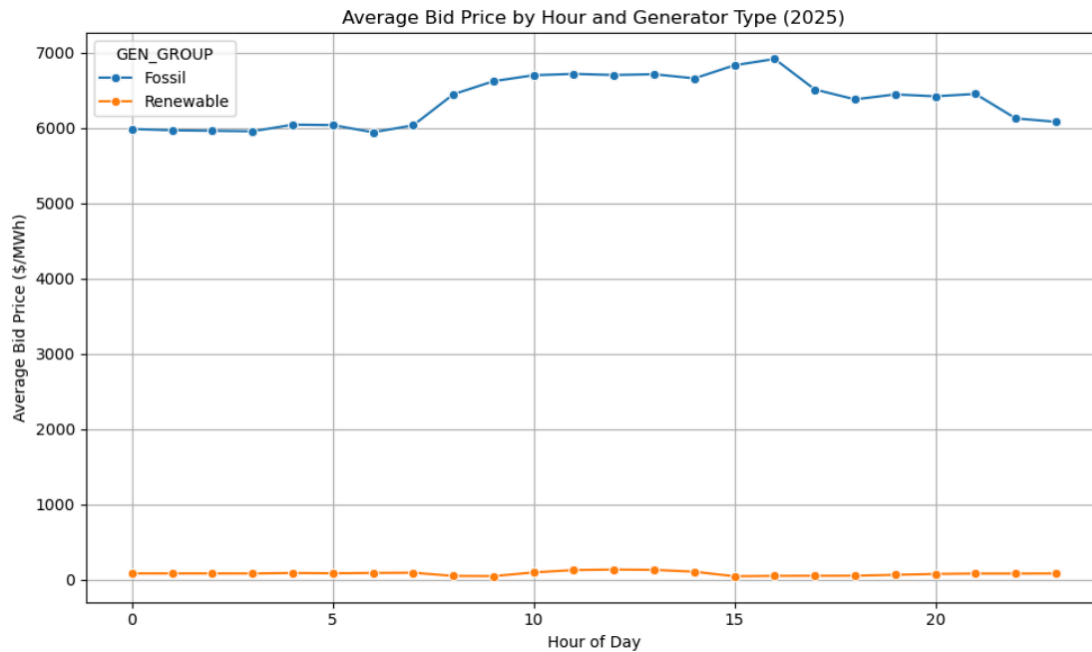


This stacked bar chart compares the **total bid volume (in MW)** submitted by each generator type across different times of the day.

- Fossil generators consistently account for a larger share of bid volume in all four time buckets. This reflects their dominant market position in terms of dispatchable capacity.
- Renewables, while still contributing significantly, offer a smaller total volume, though their share remains steady throughout the day.
- The absence of major shifts between time periods suggests that bid volume levels are stable, and any strategic behaviour is more price-driven than quantity-based.

Insight: Fossil generators not only influence market pricing through strategic bids, but also dominate the volume stack, reinforcing their role as both price setters and primary energy suppliers. Renewables play a secondary volume role, though their low-cost bids shape the lower part of the supply curve.

Hourly Price Strategy by Generator Type



This line chart presents the **average bid price by hour of day**, segmented by generator type.

- **Fossil generators** show a clear time-of-day price strategy, with bids increasing sharply during midday and peaking in the late afternoon (~Hour 16). This aligns with known demand peaks in the NEM, suggesting active revenue targeting during high-demand periods.
- **Renewables**, in contrast, display near-flat pricing across all hours, remaining close to \$0/MWh. Their bid strategy is largely non-reactive to demand conditions, consistent with their price-taker behaviour and generation constraints (e.g., solar output tied to daylight hours).

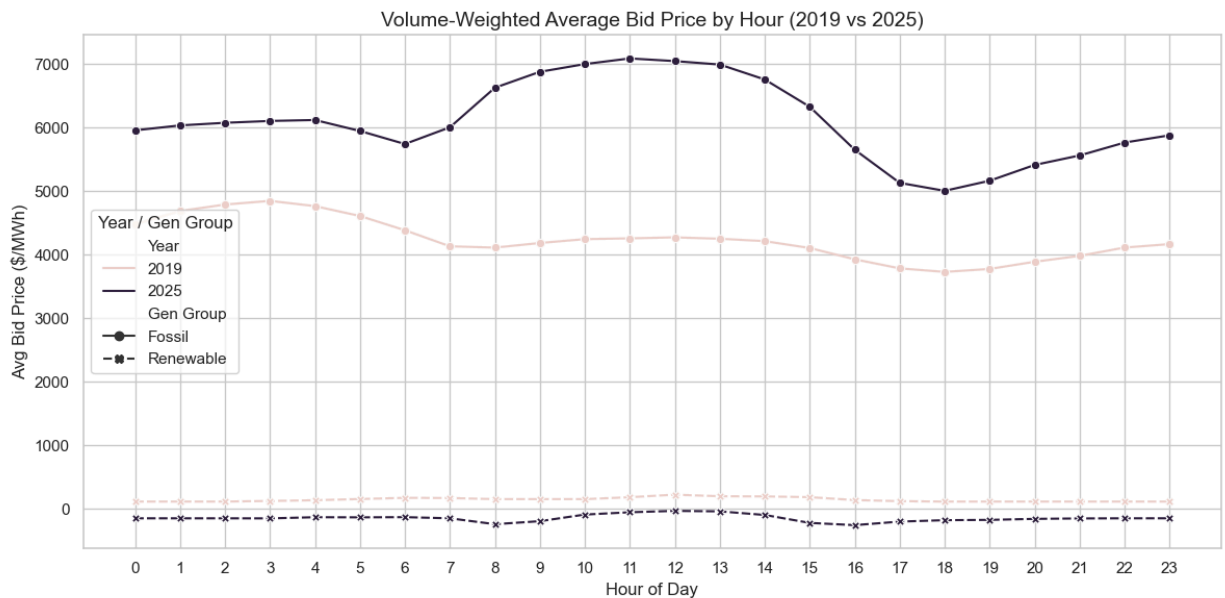
Insight: This temporal pricing pattern by fossil generators is a key strategic lever — especially relevant for investment timing and price forecasting in Task 5. The difference also sets a baseline for year-on-year comparisons in Task 4.

In summary, Task 3 has revealed clear strategic differences between renewable and fossil-fuel generators in the NEM. Fossil bids exhibit wide price flexibility and strong time-of-day pricing, while renewables remain low and stable across the day. Volume analysis confirmed the dominant market share of fossil capacity, while heatmap and hourly trends highlighted the

systematic placement of high bids. These findings form a strong foundation for the year-on-year behavioural comparison in Task 4, and will inform investment outlooks in Task 5.

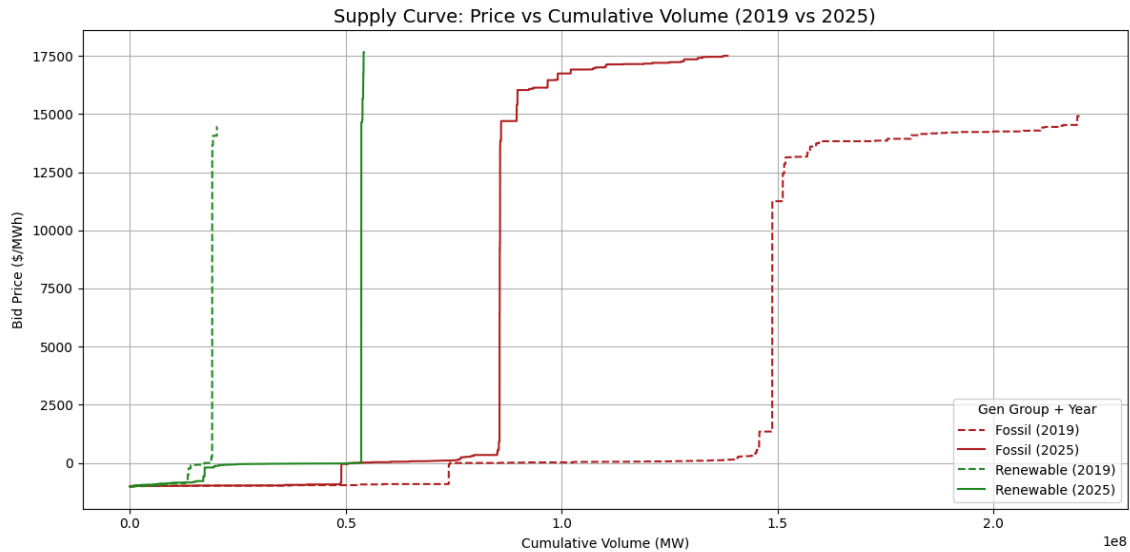
Task 4: Bidding Behaviour Comparison (2019 vs 2025)

Figure 1: Volume-Weighted Bid Price by Hour (2019 vs 2025)



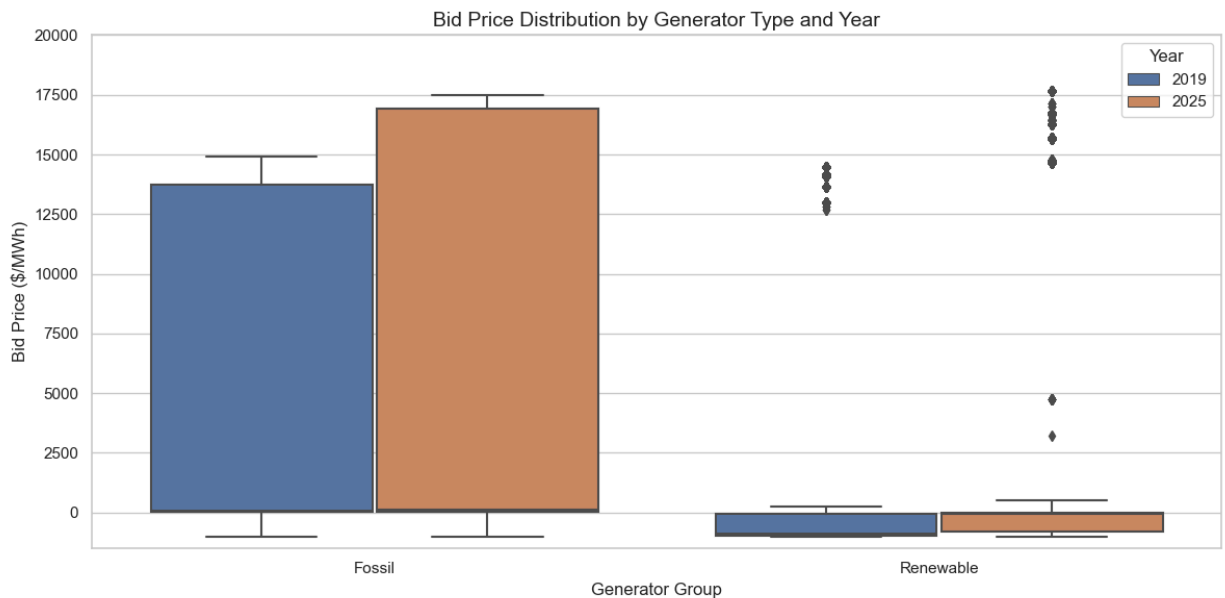
his plot illustrates the volume-weighted average bid price by hour of day across 2019 and 2025, segmented by generator group. Fossil generators show consistently high bid prices, with a noticeable spike during the midday period in 2025, suggesting more aggressive price positioning during solar troughs or high-demand hours. In contrast, renewable generators maintain negative or near-zero bid prices throughout the day in both years, reflecting their price-taking strategy driven by low marginal costs. The stark price spread between fossil and renewable bids is more pronounced in 2025, indicating a deeper market divergence in pricing strategies.

Figure 2: Supply Curve – Price vs Cumulative Volume



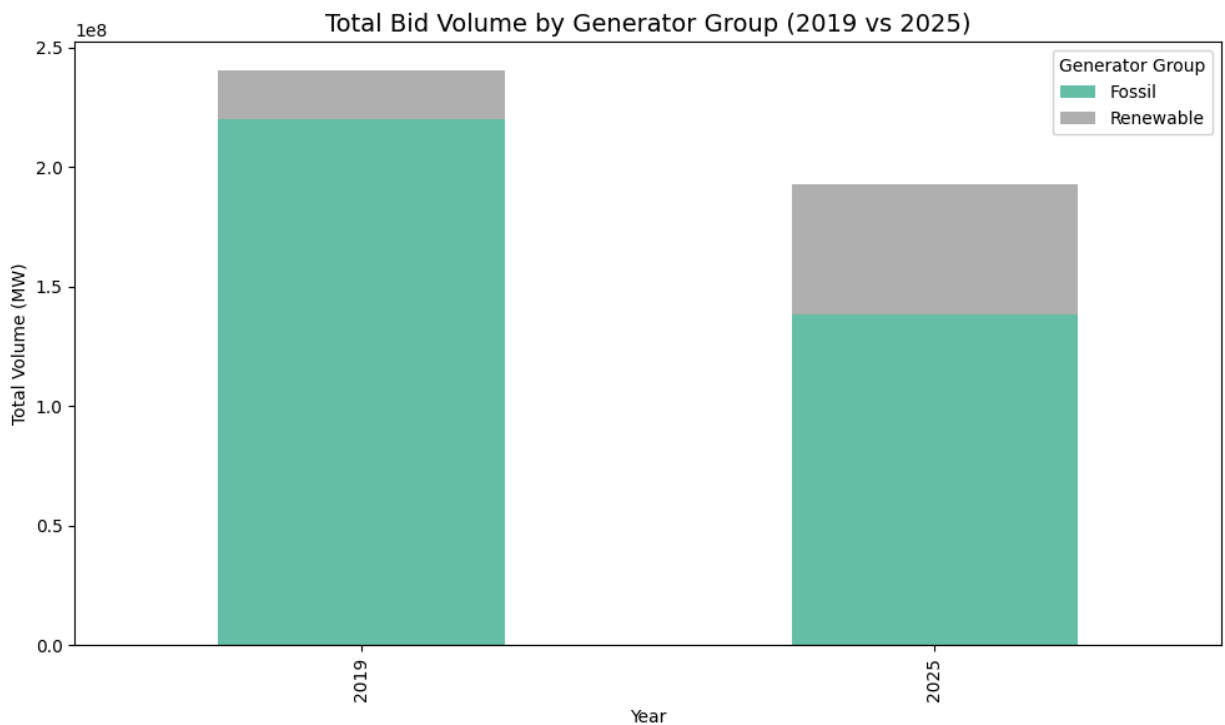
The supply curve visualises the cumulative offered volume as a function of bid price, clearly highlighting structural shifts in the market. Renewable generators in 2025 offer significantly more volume at negative prices than in 2019, indicating stronger market penetration. Conversely, the fossil supply curve in 2025 shifts upward and flattens, with broader use of high-price bids. This indicates a shift from base-load dominance toward selective, price-maximising bidding during periods of scarcity. The widening price-volume distribution reflects strategic flexibility and adaptation to a more competitive, renewables-heavy market.

Figure 3: Bid Price Distribution by Generator Type



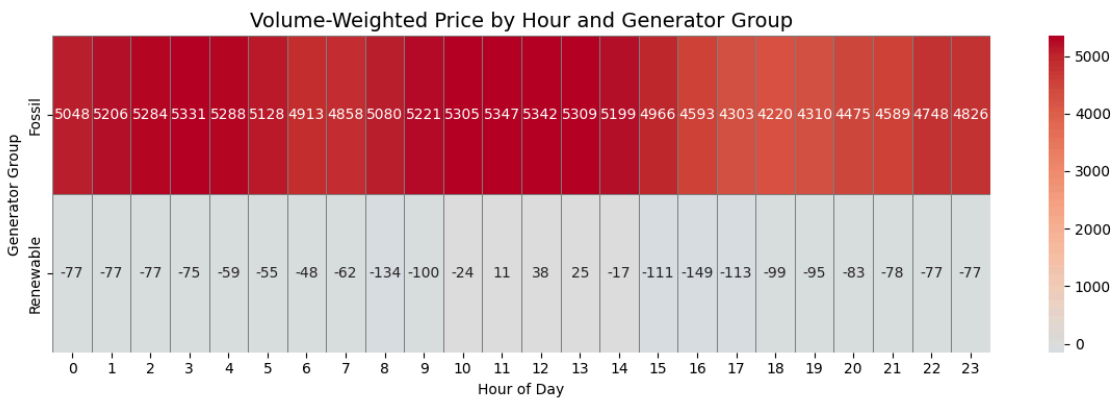
The boxplot highlights the distribution and variability of bid prices by generator group. Renewable bids remain tightly clustered around -\$1000/MWh in both 2019 and 2025, maintaining a consistent price-taker position. Fossil bids, on the other hand, display a broader distribution, with a higher median and wider interquartile range in 2025. This suggests increased volatility and a more opportunistic bidding strategy, especially in response to price volatility and renewable intermittency. The presence of more high-end outliers in 2025 further indicates fossil generators positioning for peak-period earnings.

Figure 4: Total Bid Volume by Generator Type



This chart shows a clear shift in the volume contribution of generator types between 2019 and 2025. Fossil generators, while still contributing a significant portion, show a decrease in total volume offered in 2025. In contrast, renewable generators have expanded their volume share substantially, indicating increased integration into the supply mix. This shift signals a broader market transformation — one where clean energy sources are increasingly displacing fossil assets in baseline and shoulder load supply, altering the dynamics of both pricing and capacity planning.

Figure 5: Volume-Weighted Price Heatmap by Hour and Generator Group



The heatmap reveals distinct hourly pricing strategies across generator types. Fossil generators consistently bid at elevated prices throughout the day, with peak volume-weighted prices occurring midday and late night — aligning with solar drop-off periods and possible demand surges. Renewable generators, in contrast, maintain negative bids throughout most hours, with slight deviations near midday suggesting brief market-responsive adjustments. This visual underscores the differing operational and economic models between fossil and renewable sources, with fossils leveraging price peaks and renewables flooding the market with low-cost energy.

Together, these visualisations offer a comprehensive, data-driven comparison of generator bidding behaviour between 2019 and 2025, revealing the structural transition toward renewable dominance, evolving fossil strategies, and increasingly dynamic market condition

Conclusions & Investment Recommendations

Recent Trends in the National Electricity Market (NEM)

Our findings indicate that the NEM is continuing to decarbonise, albeit at different rates across the regions. National emission intensity decreased by 7.7% between 2019 and 2025, although this use of an average masks significant variation. New South Wales achieved the best results of -21.7% evidence of a smooth transition, facilitated by successful uptake of renewables and coal-fired power station scope 2 retirements. Tasmania recorded a staggering 142.4% increase that can be described as perhaps rather anomalous considering it has the cleanest grid in absolute terms, and is influenced by variable hydroelectricity outcomes.

The intra-day patterns in NSW also suggest that the grid is maturing as during the evening period for most of the hours in the day, the emissions breakdown has grown much sharper during the evening hours. This has been achieved as a result of solar and the improved use of wind and storage at night. The differences in the hourly consumption patterns for households versus those for businesses also shows that businesses remain major contributors to the daytime emissions patterns due to their higher and mostly consistent demands.

From the outset of the market operations perspective, an analysis of the bidding data from 2025 indicates that renewable types of generators consistently bid at low or negative prices, similarly behaving as price takers, while fossil-fuel generators are behaving significantly more strategically, adjusting their bids based on the hour of the day and targeting periods of demand defined by residential peaks. Furthermore, the strategic bidding behavior is reinforced in the comparative thoughts between 2019-2025, where the fossil bids appear to have become more opportunistic during the high-price windows.

The increasing share of renewable volume in the bid stack, combined with apparent coordination of renewable type sources being deployed across states (especially after 2023), suggests a maturing and coordinated transition. If this is the trend we are observing, then it appears that we are moving from fossil baseloads toward a more dynamic, renewables-supported grid.

Investment Recommendations

i) Renewable Generation Assets (Solar/Wind):

Investors have increasing confidence in medium- to longer-term prospects for solar and wind assets. The increasing consistency in low bid prices indicates the maturity of the technologies and that they can access dispatch under all market conditions. Renewables have a lower price of realisation but provide stability and low-risk investment opportunities for investors, particularly in NSW and VIC, which provided good policy settings and were infrastructure-ready. Similarly, volume stability and increased demand for storage-backed renewables, position them for critical roles in the future capacity mix in the NEM.

ii) Gas-Fired Power Plants:

Gas-fired assets remain relevant but are changing uses to peaking and strategic roles. The broad price range and higher median bids seen in fossil bidding behaviour (particularly in the evening) indicate that gas plants are able to take advantage of volatility and short-lasting spike prices. However, the profitability of gas plants will be increasingly linked to reasonable demand forecasts and flexible operation. Investors will have to be concerned about any long-term viability as renewable technologies and storage continue to improve. Gas peaking plants, however, may still offer opportunistic returns in areas like QLTD where coal is being removed more gradually.